MAL6150: Probability and Statistics

Practice Sheet: Student Solutions Outline

Please refer this only to check your final answer, or to understand whether you are on the right track. I have not shown all the steps in these outlines!

1. $\mathbb{E}[X] = 0 \cdot \frac{1}{4} + 1 \cdot \frac{1}{2} + 2 \cdot \frac{1}{4} = \boxed{1}$.

2. $1 = c\sum_{n=1}^{\infty} 3^{-n} = \frac{c}{2} \;\Rightarrow\; c = 2; \qquad \mathbb{E}[X] = 2\sum_{n=1}^{\infty} n3^{-n} = \boxed{\frac{3}{2}}$.

3. $\mathbb{E}[X^2] = \int_0^1 x^2(2x)\,dx = \boxed{\frac{1}{2}}$.

4. $\mathbb{E}|X| = \frac{1}{3}\left(\int_{-1}^{0}(-x)\,dx + \int_0^2 x\,dx\right) = \boxed{\frac{5}{6}}$.
Keep in mind: $|x| = -x$ to the left of 0; split the integral there.

5. $\mathrm{Var}(2X-5) = 4\,\mathrm{Var}(X) = 4(11-3^2) = \boxed{8}$.
Keep in mind: Adding a constant does not change variance; multiplying by 2 multiplies it by 4.

6. $\mathbb{E}[(X-1)^2] = \frac{1}{2}\int_0^2 (x-1)^2\,dx = \boxed{\frac{1}{3}}$.

7. $\mathbb{P}(Y=0) = \mathbb{P}(X=0) = \frac{1}{2}, \qquad \mathbb{P}(Y=1) = \mathbb{P}(X=-1) + \mathbb{P}(X=1) = \frac{1}{2}$.
Keep in mind: Two distinct values of X can give the same value of X^2.

8. For $2 < y < 5$,

$$F_Y(y) = \mathbb{P}\big(X \le (y-2)/3\big) = (y-2)/3,$$

$$F_Y(y) = \begin{cases} 0, & y \le 2, \\ (y-2)/3, & 2 < y < 5, \\ 1, & y \ge 5. \end{cases} \quad (\text{all } y).$$

9. For $0 < y < 1$,

$$F_Y(y) = \mathbb{P}(-\sqrt{y} \le X \le \sqrt{y}) = \sqrt{y},$$
$$f_Y(y) = F_Y'(y) = \boxed{\tfrac{1}{2\sqrt{y}}}.$$

Outside $(0,1)$, the density is zero.
Keep in mind: Both branches $X = \pm\sqrt{Y}$ contribute to the density.

10. Write $Y = \sin\Theta$ with $\Theta \sim \mathrm{Unif}(0, 2\pi)$. For $-1 < y < 1$,

$$F_Y(y) = \frac{\pi + 2\arcsin y}{2\pi} = \boxed{\frac{1}{2} + \frac{\arcsin y}{\pi}},$$

$$f_Y(y) = F_Y'(y) = \boxed{\frac{1}{\pi\sqrt{1-y^2}}}.$$

Outside $[-1,1]$, F_Y is 0 or 1, and $f_Y = 0$.
Keep in mind: Uniform by arc-length is the same as a uniform angle at the centre.

11. $\mathbb{E}|X|^3 = 8 < \infty \;\Rightarrow\; \boxed{\lim_{n\to\infty} n^3 \mathbb{P}(|X| > n) = 0}$.

Keep in mind: This is an application of the finite-moment tail theorem, not a computation of the tail distribution.

12.

$$\begin{aligned}\mathbb{E}[X^2] &= 2\int_0^\infty \frac{t}{(1+t)^2}\,dt \\ &\geq \frac{1}{2}\int_1^\infty \frac{dt}{t} = \boxed{\infty}.\end{aligned}$$

Keep in mind: For $t \geq 1$, $t/(1+t)^2 \geq 1/(4t)$.

13.

$$\begin{aligned}\mathbb{E}[X^2] &= 2\int_0^\infty t\,\mathbb{P}(|X| > t)\,dt \\ &\leq 2\int_0^1 t\,dt + 8\int_1^\infty t^{-3}\,dt = \boxed{5}.\end{aligned}$$

Keep in mind: Use $\mathbb{P}(|X| > t) \leq 1$ for $t < 1$; the given bound applies only for $t \geq 1$.

14.

$$\mathbb{E}[X^r] = r\int_0^\infty \frac{t^{r-1}}{(1+t)^3}\,dt,$$

$$\frac{t^{r-1}}{(1+t)^3} \sim t^{r-4} \quad (t\to\infty) \implies \boxed{0 < r < 3}.$$

Keep in mind: Near 0, the integrand behaves like t^{r-1}, which is integrable for $r > 0$; at $r = 3$ the tail integral diverges.

15.

$$t^2\mathbb{P}(X > t) = \frac{(t/(1+t))^2}{\log(e+t)} \longrightarrow 0,$$

$$\mathbb{E}[X^2] = 2\int_0^\infty \frac{t\,dt}{(1+t)^2\log(e+t)} = \boxed{\infty}.$$

Keep in mind: For large t, the integrand is comparable to $1/(t\log t)$, whose integral diverges. The converse of the tail theorem fails.

16. Using $n = \sum_{k=1}^n 1$ and $n^2 = \sum_{k=1}^n (2k-1)$,

$$\mathbb{E}[X] = \sum_{k\geq 1}\mathbb{P}(X \geq k) = \sum_{k\geq 1}\frac{1}{k^2} < \infty,$$

$$\mathbb{E}[X^2] = \sum_{k\geq 1}(2k-1)\mathbb{P}(X \geq k) = \sum_{k\geq 1}\frac{2k-1}{k^2} = \boxed{\infty}.$$

Keep in mind: Interchange nonnegative sums; $(2k-1)/k^2$ behaves like $2/k$.